

## \* Assignment 2

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### I] Grade Evaluation System.

Write a java program that calculates the average marks of a student and determines the grade based on the following criteria

~~Criteria~~

Grade A :- Avg marks  $\geq 90$

Grade B :- Avg marks between 70 and 89

-||- C :- Avg marks between 50 and 69

-||- D :- Avg marks between 30 and 49

Fail :- Avg marks  $< 30$

### Predefined Values

- Marks in maths = 80
- Marks in science = 85
- Marks in History = 90

Expected o/p :- Avg marks = 85  
Grade = B

→ public class GradeEvaluation {  
 public static void main (String[] args) {

int maths = 80;

int science = 85;

int history = 90;

double avg = (maths + science + history) / 3.0;  
 char grade;

if (average >= 90)

grade = 'A';

else if (average >= 70)

grade = 'B';

else if (average >= 50)

grade = 'C';

else if (average >= 30)

grade = 'D';

else

grade = 'F';

System.out.println ("Average marks: " + (int) avg);  
 System.out.println ("Grade" + grade);

}

}

## 2) Leap year

write a java program that checks whether the year is a leap year or not. A year is a leap year if: It is divisible by 4, but not divisible by 100, or It is divisible by 400.

predefined value

Year = 2024

Year = 1900

Expected output:- 2024 is a leap year  
1900 is not a leap year.

## 3) code

```
public class LeapYear {  
    public static void main (String[] args) {
```

```
        int year1 = 2024;  
        int year2 = 1900;
```

```
        if (year1 % 4 == 0 && year1 % 100 != 0 ||  
            year1 % 400 == 0) {
```

```
            System.out.println(year1 + " is a leap year");
```

```
        } else {
```

```
            System.out.println(year1 + " is not a leap year");
```

```
        }
```

```
if (year2 % 4 == 0 + year2 % 100 != 0) ||
    year2 % 4 == 0) {
```

```
    System.out.println(year2 + "Is a leap year")
} else {
    System.out.println(year2 + "Is not a leap year")
}
```

### 3) Days of the week

write a java program that takes an integer between 1 and 7 and prints the corresponding day of the week using a switch-case statement. If the input is outside the range of 1 to 7, the program should display "Invalid day number".

predefined value:

Day number = 3

Expected output:- The day is wednesday.

```
public class DaysOfWeek {  
    public static void main (String[] args) {
```

```
        int day = 3;
```

```
        switch (day) {
```

```
            case 1:
```

```
                System.out.println ("The day is Monday");  
                break;
```

```
            case 2:
```

```
                System.out.println ("The day is Tuesday");  
                break;
```

```
            case 3:
```

```
                System.out.println ("The day is Wednesday");  
                break;
```

```
            case 4:
```

```
                System.out.println ("The day is Thursday");
```

Case 5:

```
System.out.println("The day is Friday");  
break;
```

Case 6:

```
System.out.println("The day is Saturday");  
break;
```

Case 7:

```
System.out.println("The day is Sunday");  
break;
```

default:

```
System.out.println("Invalid day number");
```

3  
3  
3

4) Identify the value of uninitialized variables

You are working on a program that handles different data types. Your manager has asked you to quickly check the values of various variables, but you're in a rush and forget to initialize them.

```
public class UninitializedVariables {  
    public static void main (String[] args) {
```

```
        byte a = 0;  
        short b = 0;  
        int c = 0;  
        long d = 0L;  
        float e = 0.0f;  
        double f = 0.0;  
        char g = "Sanket";  
        boolean h = false;
```

```
        System.out.println ("byte:" + a);  
        System.out.println ("short:" + b);  
        System.out.println ("int:" + c);  
        System.out.println ("long:" + d);  
        System.out.println ("float:" + e);  
        System.out.println ("double:" + f);  
        System.out.println ("char:" + g);  
        System.out.println ("boolean:" + h);
```

}